

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA0503

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Scenario-Based Task (High)

Assessment Tool Weightage: 35%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Given the scenario of a School Management System, identify relations and write SQL to list all students with their class teacher and grade.	BL3 – Apply	5
2	A hospital wants to track patient appointments. Design the relational schema and write SQL to find doctors with more than 10 appointments this month.	BL3 – Apply	5
3	In an e-commerce scenario, write SQL queries using sub-queries to find products that have never been ordered.	BL3 – Apply	5
4	For a university database, write relational algebra expressions to find students enrolled in both 'DBMS' and 'OS' courses.	BL3 – Apply	5
5	Given a payroll scenario, create a view showing employees whose salary is below department average and write a query to update their salary by 10%.	BL6 – Create	5
6	In a banking scenario, apply JOIN operations to retrieve customer account details, transaction history, and branch information.	BL3 – Apply	5

7	A retail store needs daily sales summary. Write SQL with GROUP BY and HAVING to report total sales per category exceeding Rs. 10,000.	BL3 – Apply	5
8	For a library system scenario, construct tuple relational calculus expressions to find members who borrowed books from all genres.	BL3 – Apply	5
9	Given an inventory management scenario, define CHECK and FOREIGN KEY constraints and demonstrate their enforcement through SQL.	BL3 – Apply	5
10	Design a relational schema for an HR system and create materialized views for monthly attendance and performance summaries.	BL6 – Create	5

Code of Conduct:

I Gudi Rama Krishna (192565028) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G Rama Krishna

RUBRICS FOR CALCULATION:

Scenario based assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. School Management System

-- Create Tables

CREATE TABLE Teacher (

```
TeacherID INT PRIMARY KEY,  
TeacherName VARCHAR(50)  
);  
  
CREATE TABLE Class (  
ClassID INT PRIMARY KEY,  
ClassName VARCHAR(20),  
TeacherID INT,  
FOREIGN KEY (TeacherID) REFERENCES Teacher(TeacherID)  
);
```

```
CREATE TABLE Student (  
StudentID INT PRIMARY KEY,  
StudentName VARCHAR(50),  
Grade CHAR(2),  
ClassID INT,  
FOREIGN KEY (ClassID) REFERENCES Class(ClassID)  
);
```

```
-- Insert Data  
INSERT INTO Teacher VALUES  
(1,'Ramesh'),  
(2,'Priya');
```

```
INSERT INTO Class VALUES  
(101,'10A',1),  
(102,'10B',2);
```

```
INSERT INTO Student VALUES  
(1,'Sahil','A',101),  
(2,'Rahul','B',102);
```

```
-- Query  
SELECT s.StudentName,c.ClassName,t.TeacherName,s.Grade  
FROM Student s  
JOIN Class c ON s.ClassID=c.ClassID  
JOIN Teacher t ON c.TeacherID=t.TeacherID;
```

OUTPUT :

```
mysql>
mysql> -- Display all students with class teacher and grade
mysql> SELECT
    ->     s.StudentID,
    ->     s.StudentName,
    ->     c.ClassName,
    ->     t.TeacherName,
    ->     s.Grade
    -> FROM Student s
    -> JOIN Class c
    -> ON s.ClassID = c.ClassID
    -> JOIN Teacher t
    -> ON c.TeacherID = t.TeacherID;
```

StudentID	StudentName	ClassName	TeacherName	Grade
1	Sahil	10A	Ramesh	A
3	Aman	10A	Ramesh	A+
2	Rahul	10B	Priya	B

```
3 rows in set (0.04 sec)
```

2. Hospital Appointment System

```
CREATE TABLE Doctor(
DoctorID INT PRIMARY KEY,
DoctorName VARCHAR(50)
);
```

```
CREATE TABLE Appointment(
AppointmentID INT PRIMARY KEY,
DoctorID INT,
AppointmentDate DATE,
FOREIGN KEY(DoctorID) REFERENCES Doctor(DoctorID)
);
```

```
SELECT d.DoctorName,COUNT(*) AS TotalAppointments
FROM Doctor d
JOIN Appointment a
ON d.DoctorID=a.DoctorID
WHERE MONTH(a.AppointmentDate)=MONTH(CURDATE())
AND YEAR(a.AppointmentDate)=YEAR(CURDATE())
GROUP BY d.DoctorID,d.DoctorName
HAVING COUNT(*)>10;
```

OUTPUT :

```
mysql> -- Query: Doctors with more than 10 appointments this month
mysql> SELECT
  ->     d.DoctorID,
  ->     d.DoctorName,
  ->     COUNT(a.AppointmentID) AS TotalAppointments
  -> FROM Doctor d
  -> JOIN Appointment a
  -> ON d.DoctorID = a.DoctorID
  -> WHERE MONTH(a.AppointmentDate) = MONTH(CURDATE())
  ->    AND YEAR(a.AppointmentDate) = YEAR(CURDATE())
  -> GROUP BY d.DoctorID, d.DoctorName
  -> HAVING COUNT(a.AppointmentID) > 10;
```

DoctorID	DoctorName	TotalAppointments
1	Dr. Kumar	11

```
1 row in set (0.06 sec)
```

3. E-Commerce Products Never Ordered

```
CREATE TABLE Product(
ProductID INT PRIMARY KEY,
ProductName VARCHAR(50)
);
```

```
CREATE TABLE Orders(
OrderID INT PRIMARY KEY,
ProductID INT,
FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
);
```

```
SELECT ProductName
FROM Product
WHERE ProductID NOT IN
(
SELECT ProductID
FROM Orders
);
```

OUTPUT :

```
mysql> -- Query: Products Never Ordered
mysql> SELECT ProductID, ProductName
    -> FROM Product
    -> WHERE ProductID NOT IN
    -> (
    ->     SELECT ProductID
    ->     FROM Orders
    -> );
```

ProductID	ProductName
103	Keyboard
104	Monitor

```
2 rows in set (0.01 sec)
```

4. Relational Algebra

-- Create Database

```
CREATE DATABASE IF NOT EXISTS UniversityDB;
```

```
USE UniversityDB;
```

-- Drop tables if they exist

```
DROP TABLE IF EXISTS Enrollment;
```

```
DROP TABLE IF EXISTS Course;
```

```
DROP TABLE IF EXISTS Student;
```

-- Create Student Table

```
CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50)
);
```

-- Create Course Table

```
CREATE TABLE Course (
    CourseID INT PRIMARY KEY,
    CourseName VARCHAR(50)
);
```

-- Create Enrollment Table

```
CREATE TABLE Enrollment (
    StudentID INT,
    CourseID INT,
    PRIMARY KEY (StudentID, CourseID),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);
```

-- Insert Sample Data

INSERT INTO Student VALUES

(1,'Sahil'),

(2,'Rahul'),

(3,'Aman');

INSERT INTO Course VALUES

(101,'DBMS'),

(102,'OS'),

(103,'Java');

INSERT INTO Enrollment VALUES

(1,101),

(1,102),

(2,101),

(3,102),

(3,103);

-- Students enrolled in BOTH DBMS and OS

SELECT s.StudentID, s.StudentName

FROM Student s

JOIN Enrollment e ON s.StudentID = e.StudentID

JOIN Course c ON e.CourseID = c.CourseID

WHERE c.CourseName IN ('DBMS','OS')

GROUP BY s.StudentID, s.StudentName

HAVING COUNT(DISTINCT c.CourseName) = 2;

OUTPUT :

```
mysql> SELECT s.StudentID,
->         s.StudentName
-> FROM Student s
-> JOIN Enrollment e
-> ON s.StudentID = e.StudentID
-> JOIN Course c
-> ON e.CourseID = c.CourseID
-> WHERE c.CourseName IN ('DBMS','OS')
-> GROUP BY s.StudentID, s.StudentName
-> HAVING COUNT(DISTINCT c.CourseName) = 2;
+-----+-----+
| StudentID | StudentName |
+-----+-----+
|          1 | Sahil       |
+-----+-----+
1 row in set (0.03 sec)
```

5. Payroll View

CREATE TABLE Employee(


```

EmpID INT PRIMARY KEY,
EmpName VARCHAR(50),
DepartmentID INT,
Salary DECIMAL(10,2)
);

```

```

CREATE VIEW LowSalaryEmployees AS
SELECT *
FROM Employee e
WHERE Salary <
(
SELECT AVG(Salary)
FROM Employee
WHERE DepartmentID=e.DepartmentID
);

```

```

UPDATE Employee
SET Salary=Salary*1.10
WHERE EmpID IN
(
SELECT EmpID
FROM LowSalaryEmployees
);

```

OUTPUT :

```

mysql> -- Display Updated Employee Table
mysql> SELECT * FROM Employee;
+-----+-----+-----+-----+
| EmpID | EmpName | DepartmentID | Salary |
+-----+-----+-----+-----+
| 1 | Sahil | 101 | 25000.00 |
| 2 | Rahul | 101 | 35000.00 |
| 3 | Aman | 102 | 20000.00 |
| 4 | Priya | 102 | 30000.00 |
| 5 | Neha | 103 | 40000.00 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

6. Banking JOIN

```

CREATE TABLE Customer(
CustomerID INT PRIMARY KEY,
CustomerName VARCHAR(50)
);

```

```

CREATE TABLE Branch(
BranchID INT PRIMARY KEY,
BranchName VARCHAR(50)
);

```

);

```
CREATE TABLE Account(  
  AccountID INT PRIMARY KEY,  
  CustomerID INT,  
  BranchID INT,  
  FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID),  
  FOREIGN KEY(BranchID) REFERENCES Branch(BranchID)  
);
```

```
CREATE TABLE Transactions(  
  TransactionID INT PRIMARY KEY,  
  AccountID INT,  
  Amount DECIMAL(10,2),  
  FOREIGN KEY(AccountID) REFERENCES Account(AccountID)  
);
```

```
SELECT c.CustomerName,  
       a.AccountID,  
       t.TransactionID,  
       t.Amount,  
       b.BranchName  
FROM Customer c  
JOIN Account a ON c.CustomerID=a.CustomerID  
JOIN Transactions t ON a.AccountID=t.AccountID  
JOIN Branch b ON a.BranchID=b.BranchID;
```

OUTPUT :

```
-> t.TransactionID,  
-> t.TransactionDate,  
-> t.TransactionType,  
-> t.Amount,  
-> a.Balance  
-> FROM Customer c  
-> JOIN Account a  
-> ON c.CustomerID = a.CustomerID  
-> JOIN Branch b  
-> ON a.BranchID = b.BranchID  
-> JOIN Transactions t  
-> ON a.AccountID = t.AccountID;
```

CustomerID	CustomerName	AccountID	BranchName	BranchCity	TransactionID	TransactionDate	TransactionType	Amount	Balance
1	Sahil	10000.00	50000.00	1001	Chennai Branch	Chennai	1	2026-08-01	Deposit
1	Sahil	5000.00	50000.00	1001	Chennai Branch	Chennai	2	2026-08-02	Withdrawal
2	Rahul	7000.00	30000.00	1002	Bangalore Branch	Bangalore	3	2026-08-03	Deposit

3 rows in set (0.00 sec)

7. Retail Store GROUP BY

```
CREATE TABLE Sales(  
  ProductID INT PRIMARY KEY,  
  CustomerID INT,  
  BranchID INT,  
  FOREIGN KEY(ProductID) REFERENCES Product(ProductID),  
  FOREIGN KEY(BranchID) REFERENCES Branch(BranchID)  
);
```

```

SaleID INT PRIMARY KEY,
Category VARCHAR(50),
Amount DECIMAL(10,2)
);

```

```

INSERT INTO Sales VALUES
(1,'Electronics',6000),
(2,'Electronics',7000),
(3,'Clothing',4000),
(4,'Clothing',3000),
(5,'Furniture',12000);

```

```

SELECT Category,
SUM(Amount) AS TotalSales
FROM Sales
GROUP BY Category
HAVING SUM(Amount)>10000;

```

OUTPUT :

```

mysql> -- Query: Total sales per category exceeding Rs. 10,000
mysql> SELECT
->     Category,
->     SUM(Amount) AS TotalSales
-> FROM Sales
-> GROUP BY Category
-> HAVING SUM(Amount) > 10000;
+-----+-----+
| Category | TotalSales |
+-----+-----+
| Electronics | 13000.00 |
| Furniture | 12000.00 |
+-----+-----+
2 rows in set (0.00 sec)

```

8. Tuple Relational Calculus

```

-- Create Database
CREATE DATABASE IF NOT EXISTS LibraryDB;
USE LibraryDB;

```

```

-- Drop tables if they exist
DROP TABLE IF EXISTS Borrow;
DROP TABLE IF EXISTS Book;
DROP TABLE IF EXISTS Member;
DROP TABLE IF EXISTS Genre;

```

```

-- Create Tables

```

```
CREATE TABLE Member (  
    MemberID INT PRIMARY KEY,  
    MemberName VARCHAR(50)  
);
```

```
CREATE TABLE Genre (  
    GenreID INT PRIMARY KEY,  
    GenreName VARCHAR(50)  
);
```

```
CREATE TABLE Book (  
    BookID INT PRIMARY KEY,  
    BookName VARCHAR(50),  
    GenreID INT,  
    FOREIGN KEY (GenreID) REFERENCES Genre(GenreID)  
);
```

```
CREATE TABLE Borrow (  
    MemberID INT,  
    BookID INT,  
    PRIMARY KEY (MemberID, BookID),  
    FOREIGN KEY (MemberID) REFERENCES Member(MemberID),  
    FOREIGN KEY (BookID) REFERENCES Book(BookID)  
);
```

```
-- Insert Sample Data  
INSERT INTO Member VALUES  
(1,'Sahil'),  
(2,'Rahul');
```

```
INSERT INTO Genre VALUES  
(1,'DBMS'),  
(2,'Programming'),  
(3,'Networks');
```

```
INSERT INTO Book VALUES  
(101,'Database Systems',1),  
(102,'C Programming',2),  
(103,'Computer Networks',3);
```

```
INSERT INTO Borrow VALUES  
(1,101),  
(1,102),  
(1,103),
```

(2,101);

-- Members who borrowed books from ALL genres

```
SELECT m.MemberID, m.MemberName
FROM Member m
JOIN Borrow b ON m.MemberID = b.MemberID
JOIN Book bk ON b.BookID = bk.BookID
GROUP BY m.MemberID, m.MemberName
HAVING COUNT(DISTINCT bk.GenreID) = (SELECT COUNT(*) FROM Genre);
```

OUTPUT :

```
mysql> SELECT
->     m.MemberID,
->     m.MemberName
-> FROM Member m
-> JOIN Borrow br
-> ON m.MemberID = br.MemberID
-> JOIN Book b
-> ON br.BookID = b.BookID
-> GROUP BY m.MemberID, m.MemberName
-> HAVING COUNT(DISTINCT b.GenreID) = (
->     SELECT COUNT(*) FROM Genre
-> );
```

MemberID	MemberName
1	Sahil

now in set (0.05 sec)

9. Inventory Constraints

```
CREATE TABLE Supplier(
SupplierID INT PRIMARY KEY,
SupplierName VARCHAR(50)
);
```

```
CREATE TABLE Product(
ProductID INT PRIMARY KEY,
ProductName VARCHAR(50),
Price DECIMAL(10,2) CHECK(Price>0),
SupplierID INT,
FOREIGN KEY(SupplierID)
REFERENCES Supplier(SupplierID)
);
```

```
INSERT INTO Supplier VALUES
(1,'ABC Suppliers');
```

```
INSERT INTO Product VALUES
(101,'Keyboard',800,1);
```

```
-- Invalid (CHECK fails)
```

```
-- INSERT INTO Product VALUES(102,'Mouse',-500,1);
```

```
-- Invalid (FOREIGN KEY fails)
```

```
-- INSERT INTO Product VALUES(103,'Monitor',7000,5);
```

OUTPUT :

```
mysql>
mysql> -- Display Data
mysql> SELECT * FROM Supplier;
+-----+-----+
| SupplierID | SupplierName |
+-----+-----+
|          1 | ABC Suppliers |
|          2 | XYZ Suppliers |
+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Product;
+-----+-----+-----+-----+-----+
| ProductID | ProductName | Price | Quantity | SupplierID |
+-----+-----+-----+-----+-----+
|          101 | Keyboard | 800.00 | 20 | 1 |
|          102 | Mouse | 500.00 | 50 | 2 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

10. HR System (Materialized View Equivalent in MySQL)

```
CREATE TABLE Employee(
EmpID INT PRIMARY KEY,
EmpName VARCHAR(50)
);
```

```
CREATE TABLE Attendance(
AttendanceID INT PRIMARY KEY,
EmpID INT,
AttendanceDate DATE,
Status VARCHAR(10),
FOREIGN KEY(EmpID) REFERENCES Employee(EmpID)
);
```

```
CREATE TABLE Performance(
PerformanceID INT PRIMARY KEY,
EmpID INT,
Score INT,
FOREIGN KEY(EmpID) REFERENCES Employee(EmpID)
```

);

-- Monthly Attendance Summary

```
CREATE TABLE MonthlyAttendance AS
SELECT EmpID,
MONTH(AttendanceDate) AS Month,
COUNT(*) AS TotalDaysPresent
FROM Attendance
WHERE Status='Present'
GROUP BY EmpID,MONTH(AttendanceDate);
```

-- Monthly Performance Summary

```
CREATE TABLE MonthlyPerformance AS
SELECT EmpID,
AVG(Score) AS AverageScore
FROM Performance
GROUP BY EmpID;
```

SELECT * FROM MonthlyAttendance;

SELECT * FROM MonthlyPerformance;

OUTPUT :

```
mysql>
mysql> -- Display Summary Tables
mysql> SELECT * FROM MonthlyAttendance;
+-----+-----+-----+
| EmpID | Month | TotalAttendance |
+-----+-----+-----+
|      1 |      8 |                2 |
|      2 |      8 |                1 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM MonthlyPerformance;
+-----+-----+
| EmpID | AverageScore |
+-----+-----+
|      1 |      87.5000 |
|      2 |      75.0000 |
|      3 |      80.0000 |
+-----+-----+
3 rows in set (0.00 sec)
```